

Do Hand-Engineered Features Still Help? Feature Richness Versus Data Scale in Supervised 7×7 Go Move Prediction

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Abstract—This paper studies whether hand-engineered Go feature planes (stone liberties, move history) still improve supervised move prediction as labelled data volume grows, on a controlled 7×7 Go corpus generated by KataGo self-play. A 3 (feature richness: $N = 2, 4, 7$ planes) $\times$ 4 (data volume) factorial compares classical baselines (logistic regression, random forest, k -NN, linear SVM) against a convolutional policy/value network. Feature richness and architecture interact rather than combine additively: the same planes worth +0.11 top-1 accuracy to a linear model are worth 0.0008 — below seed noise — to the CNN, which reaches 90.4% top-1 / 98.7% top-3 unaided on a sealed test split (mean over 5 seeds, read exactly once) and overtakes every classical baseline once training data exceeds ~ 13 k unique positions. A 150-game round-robin further shows that PUCT search over the same frozen weights adds strength (≈ 400 Elo, $p=4.1\times 10^{-3}$) substantially exceeding any single supervised-learning design choice in the study. All data, code, and analysis scripts are publicly available at [repository URL withheld pending release].

Index Terms—Go, supervised learning, feature engineering, data scarcity, convolutional neural networks, ablation study

I. INTRODUCTION

As labelled self-play data grows, does hand-engineered structure (liberties, move history) still earn its keep against a model that can in principle learn the same structure from raw stone positions given enough data? This paper’s contributions are:

- 1) A controlled 3×4 factorial ablation isolating feature richness from data volume on a single, oracle-verified 7×7 Go corpus (120,000 KataGo self-play games).
- 2) A comparison of four classical baselines and a CNN policy/value network, compute-matched, with a properly-derived baseline for every metric (majority-class/uniform-random for policy, side-to-move for value) — a step several move-prediction papers skip.
- 3) An honest accounting of a small-board corpus’s information ceiling: 56,798 unique positions out of 3.3M rows, and what that means for a data-volume factor.
- 4) A compute-cheap unsupervised sub-study (§V-G): an autoencoder, classical clustering, and control-paired linear probes ask whether reconstruction alone recovers the same board structure supervision does.

Related work is reviewed in §II, the corpus and experimental design in §III–IV, results in §V, and their implications in §VI.

II. RELATED WORK

Search-guided self-play for Go and chess. AlphaGo [1] combined supervised policy pretraining with MCTS and RL; AlphaGo Zero [2] and AlphaZero [3] removed human data entirely. KataGo [4] is the methodological ancestor of this paper’s data pipeline and the source of its labels (§III); its ablations on auxiliary targets and input features are the closest precedent for a controlled feature-richness study, at much larger scale and board size. Kocsis and Szepesvári’s UCT [5] underlies the tree search above, and MCTS feature-biasing work [6] is the closest general-game-playing analogue of Factor A: linear feature weights instead of a full network.

Supervised pattern learning without search. DeepChess [7] is the nearest analogue for this paper’s supervised-only framing: an end-to-end network trained purely on positions and outcomes. Interpretability work on search-free and search-trained game engines — concept probing in AlphaZero [8], representation quality in chess vision transformers [9], and the control-task methodology for validating that a probe measures signal rather than capacity [10] — motivates this paper’s insistence on properly-derived baselines (§IV): a probe or classifier beating a carelessly-chosen baseline is the same failure mode in either literature.

Small-board Go and classical feature selection. Small-board Go has an established solved-game literature [11], grounding 7×7 as tractable for full self-play yet still combinatorially non-trivial. Feature richness sits in classical feature-selection theory [12] and pattern-recognition grounding [13]: features help only when a model cannot already infer their content from raw input, exactly Factor B’s question.

III. DATASET

A. Generation

The corpus is 120,000 self-play games generated by KataGo v1.16.5 (Eigen/CPU build) via its `match` subcommand, at `maxVisits = 16`, single search thread, network `g170e-b10c128-s1141046784-d204142634`. Rules are fixed and identical in generation and replay: board 7×7 , komi 9.5, positional superko, area (Tromp–Taylor) scoring, multi-stone suicide illegal; zero games were dropped for a rule or board-size mismatch.

TABLE I

TRAIN/VAL/TEST SPLIT. ALL THREE CARVED FROM ONE GENERATION RUN; GAME-LEVEL OVERLAP VERIFIED = 0 FOR EVERY PAIR.

Split	Games	Positions	Share
Train	107,969	3,310,156	90.0%
Val	6,027	184,834	5.0%
Test (sealed)	6,004	182,102	5.0%

B. Splits

All three splits (Table I) are carved from this single 120,000-game run and partitioned *by game* (never by position) via `crc32(file:line) mod 20`, which is what guarantees zero game overlap between splits — splitting by position would leak, since positions within a game are highly correlated.

The test split is touched exactly once, for the final results table; every tuning and model-selection decision in this paper uses the validation split.

C. Policy label

The task is 50-way classification (49 board points + pass). Contrary to an earlier design assumption favouring centre/tengen dominance, the board-point distribution is comparatively flat: the most frequent point (centre, 3.29%) is only 4.4× the least frequent (0.75%). The one genuinely outsized class is *pass* itself, the training-set mode at 9.09% (Fig. 1). Macro-F1 is reported throughout on this basis — “50-way classification whose largest class is *pass*,” not severe imbalance.

D. Value label: a baseline trap

White wins 90.6% of games at the game level — not a defect: removing komi from final scores shows 71% of games settle at exactly margin +9, the established solved value of 7×7 Go under perfect play. Komi 9.5 (half a point above 9, so no game draws) then hands those games to White by the minimum margin. This is a spike in the outcome distribution, not a spread, so no komi value rebalances it without changing which side wins by construction; 9.5 is kept and the skew is reported, not engineered away.

The consequence for evaluation is a value-label baseline trap. Since `values` is stored from the side-to-move’s perspective and side alternates every ply, row-level balance looks fair (train: 49.4% win / 50.6% loss) while a much stronger regularity holds at the level that matters: predicting a win iff White is to move scores 85.84% (train) / 85.55% (val) / 86.44% (test) accuracy alone. Scoring against the 50.6% marginal-majority baseline instead would overstate any result by roughly 35 points. Every value accuracy here is quoted beside the 85.55% validation-split side-to-move baseline for this reason (see also §IV for the consequence this has for the value *target*, not just the baseline).

E. Duplication and train/val position overlap

Two properties of the corpus would otherwise surface as reviewer objections and are disclosed here directly. First, the corpus is small in *unique* positions relative to its row count:

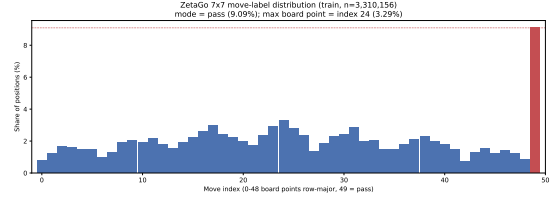


Fig. 1. Move-label distribution, training split ($n = 3,310,156$). Mode is *pass* (9.09%), not a board point; board-point distribution is comparatively flat (max 3.29%, centre). Basis for macro-F1 as “50-way classification whose largest class is *pass*,” not severe imbalance.

the full training split holds 3,310,156 rows but only 56,798 unique board states, a $58.3\times$ duplication factor, concentrated almost entirely in the opening and not a corpus-wide defect: plies 0–6 supply 22.8% of all rows but only 93 of 56,798 unique positions (0.2%, each repeated roughly 8,100 times), while plies 20 and beyond supply 91.7% of unique positions at roughly $22\times$ repetition — in unique-position terms the corpus is 92% midgame and endgame content. Training on raw rows would let the 93 opening boards dominate roughly a quarter of every gradient step, diluting any measured feature-richness effect (liberty and history planes carry almost no information on a near-empty board); §IV’s featurisation is therefore always evaluated at natural, non-duplicated row weighting, an asymmetry the reader should weigh at the smallest data-volume levels.

Second, 90.7% of validation rows hold a board position that also occurs in train. This is *not* leakage — game-level overlap is exactly zero, verified directly — but the ordinary structure of a shared opening book: every game begins from the same first move (only 17 distinct positions by ply 4), so overlap starts at 100% for ply < 10 and decays monotonically as the tree branches, to 90.2% by ply 20–30 and 50.4% by ply 50+. Opening-move determinism (not addressed here; the tested-safe remedy is randomising the first plies before self-play) is the underlying cause of both this and the row-weighting skew above.

IV. METHODOLOGY

We formally define the two experimental factors, the model set, the prediction targets, and the evaluation protocol below.

- **Factor A (feature richness).** $N = 2$: current-player and opponent stone planes. $N = 4$: $N = 2$ + normalised group liberties + turn indicator. $N = 7$: $N = 4$ + signed history at $t - 1/t - 2$ (both players’ stones, not just the mover’s own) + an occupancy-change plane.
- **Factor B (data volume).** Nested by-game nested sub-samples: 1,000 / 5,000 / 20,000 / 107,969 (full) games, corresponding to 5,501 / 12,748 / 24,277 / 56,798 unique positions — state both quantities; reporting rows alone misrepresents information content by $\sim 58\times$.
- **Models.** Logistic regression, random forest, k -NN, and linear SVM (flattened, all four compute-matched at 80k rows), plus a three-layer convolutional policy/value network on the full split.

- **Value target.** Score margin (regression), not win/loss: §III-D shows win/loss is near-solved (side-to-move baseline 85.84% train / 85.55% val) while the same rule explains only 8.1% of margin variance, so margin is the non-degenerate target our value head regresses — the same quantity KataGo’s own value head predicts. MSE floors any margin model must clear: 265.73 (predict 0), 244.09 (best constant per side-to-move). Win/loss is still reported as the derived sign of the predicted margin, beside the 85.55% baseline.
- **Baselines.** Majority-class and uniform-random-over-legal for policy; side-to-move (85.55% val) for value, not the marginal-majority baseline (50.6%), which would overstate any result by ~ 35 points.
- **Seeds and CIs.** 5 seeds per cell; 95% bootstrap confidence intervals resampled over *games*, not positions, since positions within a game are correlated.

A. Search-augmented evaluation

Beyond the supervised factorial, we evaluate whether search over a frozen, already-trained model improves playing strength relative to a one-shot policy read. We use PUCT [1], [5], selecting by $Q + c_{\text{puct}} \cdot P \cdot \sqrt{N_{\text{parent}}}/(1 + N_{\text{child}})$ down to an unexpanded leaf, evaluated once by the frozen network (policy head as prior, masked to legal moves and renormalised; value head’s margin as leaf value, negated when propagated to the opponent’s node); terminal nodes use the true Tromp–Taylor outcome. The root’s move is the highest visit count after $N=100$ simulations at $c_{\text{puct}}=1.5$ (configuration, not tuned). The greedy baseline’s pass rule (a move-history minimum before pass is eligible, since `pass` is the modal training label) applies at every search node, evaluated at that node’s own ply.

Search here is strictly *evaluation-time*: no weight update, gradient step, or reward signal occurs, and no simulated game is ever recorded as training data — the design boundary separating this from AlphaZero-style self-play, which would retrain on search-generated targets. No Dirichlet root noise is applied, since that is a training-time exploration device with no role in a pure strongest-move search.

Three agent tiers share one trained model: uniform-random over legal moves; one-shot greedy (argmax, same pass rule); and the PUCT agent above. Greedy and search wrap *identical* weights, so any gap between them is attributable to search alone (Section V). Agents meet in a round-robin with equal games per colour per pair — mandatory, since §III-D shows colour explains more of a game’s outcome here than either agent’s skill. Strength is a Bradley–Terry Elo rating with bootstrap CIs, anchored at random, alongside head-to-head records: an anchor swept without a loss drives its own strength to zero and every Elo to $+\infty$, so head-to-head is reported alongside Elo, not instead of it.

Both greedy and search are deterministic given a fixed model, so repeated games between the same pair are not independent trials unless the opening varies. We prepend a small number of uniformly random legal moves before either agent takes control, drawn independently per repetition and

TABLE II
TEST-SET RESULTS, FULL VOLUME, $N=4$, 5 SEEDS (MEAN $\pm$ SD), READ EXACTLY ONCE AFTER ALL DECISIONS WERE FROZEN ON VALIDATION.

Model	Top-1	Top-3	Macro-F1
logreg	0.7963 $\pm$ 0.0017	0.9196	0.7952
rf	0.8752 $\pm$ 0.0003	0.9600	0.8826
knn	0.8137 $\pm$ 0.0030	0.9364	0.8110
svm	0.7912 $\pm$ 0.0014	0.9157	0.7878
cnn	0.9037 $\pm$ 0.0005	0.9872	0.9158

shared between both colour assignments of that repetition, preserving colour balance position-for-position.

B. Unsupervised representation

Separately, we ask whether a convolutional autoencoder trained on raw stone positions alone recovers board structure comparable to the supervised CNN’s, and whether its weights transfer [14]. The encoder is the champion CNN’s body verbatim (three 3×3 same-padded convolutions, 64 channels, so weights are transplantable, §V-G), projected to a latent of 64 (dimension-matched to the value head’s own penultimate layer) or, secondarily, 32; the decoder mirrors this in reverse, trained with BCE-with-logits on the raw $N=2$ stone-occupancy planes. It never receives `moves`, `values`, or `margins` (a runtime assertion, not convention) and always trains deduplicated (§V-F’s `dedup=unique`, here mandatory: 56,798 unique positions, 10% held out for reconstruction-loss model selection only). For each board statistic (stone count, move number, largest/atari group counts, mean distance from centre) we fit a linear probe from a 64-d latent to that statistic, paired with a control trained identically against a row-shuffled target [10]: a probe that does not clear its control demonstrates probe capacity, not structure. *k*-means (silhouette-selected *k*) and DBSCAN on a PCA-reduced space (90%-variance components; ε from a fixed nearest-neighbour heuristic, never tuned post hoc) cluster encoded validation positions, with stability checked by adjusted Rand index across 3 autoencoder seeds. The identical protocol then runs, unchanged, on the champion CNN’s own 64-d value-head penultimate layer.

V. RESULTS

A. Move prediction

The headline number is on the sealed test split, read exactly once, after every design and model-selection decision below was frozen on validation. The full-volume $N=4$ cell was retrained at 5 seeds for this pass (distinct from §V-E/§V-G’s persisted champion — GPU/CPU nondeterminism, §VI); its seed-46 instance scores 90.46%/98.72%/91.70% (top-1/top-3/macro-F1) on test against 90.40%/98.72%/91.60% on validation, the same weights — no generalisation gap. Table II gives all five models on test; every other result (Table III onward) is validation, since Factor A/B and the tournament were run once, on validation, by design.

An exact McNemar test (CNN vs. RF, seed 46) confirms the crossover below on identical positions: of 8,847 test

TABLE III
MEAN TOP-1 ACCURACY BY MODEL AND DATA VOLUME (5 SEEDS, 3 ENCODINGS). BOLD MARKS THE ROW MAXIMUM.

Model	1k	5k	20k	full (108k)
logreg	0.7711	0.7899	0.7899	0.7901
rf	0.8589	0.8775	0.8759	0.8760
knn	0.7676	0.8123	0.8120	0.8143
svm	0.7726	0.7798	0.7802	0.7795
cnn	0.8192	0.8764	0.8934	0.9036

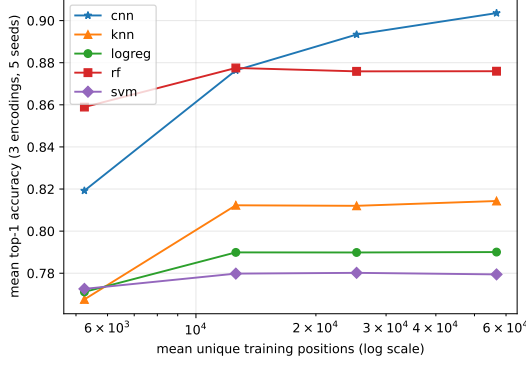


Fig. 2. Mean top-1 accuracy vs. mean unique training positions (log scale), one line per model, mean over 3 encodings and 5 seeds. Classical models plateau by $\sim 13k$ unique positions; the CNN does not.

disagreements, CNN is right and RF wrong on 7,130, the reverse on 1,717 ($p < 10^{-300}$); the same asymmetry holds on validation (6,986 vs. 1,839 of 8,825, $p < 10^{-300}$) — not a validation-set artefact. Table III reports mean top-1 across seeds/encodings for every (model, volume) cell on validation.

Every classical model captures essentially its full gain between 1k and 5k games and is flat thereafter; the CNN keeps improving to the last data point. Fig. 2 plots the same result against *unique* training positions rather than games sampled — the axis that actually varies information content, since $108 \times$ more games buys only $10.9 \times$ more unique positions (§III-E). Random forest leads below $\sim 13k$ unique positions; the CNN overtakes it once unique coverage passes that point and the gap widens monotonically. Checked per seed rather than on pooled means, all five agree on the ranking at 20k and full volume (wide, consistent CNN margin); at 5k RF leads on 4/5 seeds but by at most 0.004 — an order of magnitude below its unambiguous 0.040 lead at 1k — so we read 5k as the crossover’s near-boundary, with the CNN advantage dominating only from 20k onward.

B. Feature richness (Factor A)

Comparing each model’s $N=2 \rightarrow N=7$ spread in top-1 at full volume against its own across-seed standard deviation isolates whether feature richness helps at all, and for which models. SVM moves by 0.1095 ($70 \times$ its seed SD of 0.00156); logistic regression by 0.0900 ($77 \times$ its SD of 0.00117). Random forest moves by only 0.0021 (above noise but small); k -NN’s 0.0032 spread sits inside its own SD. The CNN’s

TABLE IV
ROUND-ROBIN STRENGTH, 25 GAMES/COLOUR/PAIR, 4 RANDOM OPENING PLIES. ELO ANCHORED AT RANDOM.

Agent	Win rate	Elo	95% CI
mcts (100 sim)	0.950	802.8	[633.0, 1161.2]
greedy (argmax)	0.500	401.4	[268.5, 588.3]
random	0.050	0.0	—

spread is 0.0008 — *below* its seed SD of 0.00081, and below a quarter of the mean bootstrap confidence-interval width on any single one of these rows (0.0032). The identical featuriser, data, and protocol that moved SVM by $145 \times$ the CNN’s spread confirms this null is a property of the model, not an underpowered experiment: liberty counts and history planes are worth roughly $+0.10$ top-1 to a linear model and nothing measurable to the convolutional network, which apparently recovers the same information from raw stone positions through its own early layers. A two-way ANOVA on top-1 (factors: encoding, model; full volume, 5 seeds) confirms this formally: the encoding $\times$ model interaction term is significant at $F(8,60)=1295$, $p < 10^{-60}$ — feature richness’s effect genuinely depends on which model receives it, not an artefact of comparing effect sizes to noise floors informally.

C. Value target

The CNN has the lowest margin MSE (69.78 vs. RF’s 72.31, full volume) but RF has the higher win/loss sign accuracy (93.17% vs. 91.71%, baseline 85.55%) — the same pattern replicates on test (§V: RF 0.9343 vs. CNN 0.9206). Different quantities: the CNN fits its regression target more precisely overall, RF is more often correct about which side of zero the margin falls on; a model can lead on one and trail the other, so both are reported.

D. Error analysis

A literal 50×50 confusion matrix is unreadable and almost entirely zeros; Fig. 3 instead plots its diagonal — per-true-point top-1 recall — on board geometry, more informative for a spatial task. The centre (index 24) recalls at 99.9%; its four orthogonal neighbours are the worst points on the board (as low as 59.0%), and the dominant confusions are between centre and these neighbours (the single most common error, 0.78% of rows, is true =(row 3, col 2) predicted (row 2, col 3), both one step from centre). *pass* (50th class) recalls at 93.0% ($n=16,849$). Consistent with the 8.3-point top-1/top-3 gap: most residual error is not confusing strong moves with weak, but ranking among several similarly strong central points, several simultaneously reasonable in 7×7 play.

E. Search-augmented playing strength

The greedy and search agents (§IV-A) wrap the identical champion weights, so any strength gap is attributable to search alone. Table IV reports the 150-game round-robin.

Head-to-head: mcts defeats greedy 45–5, Elo CIs non-overlapping. The 50 games per pair are 25 openings played

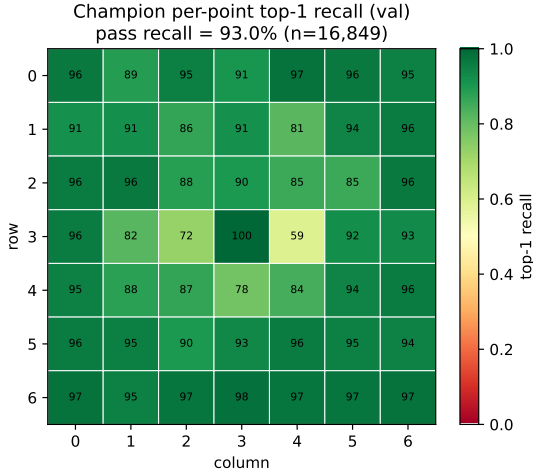


Fig. 3. Per-point top-1 recall, champion model (val) — diagonal of the 50-class confusion matrix on board geometry. `pass` (93.0% recall, $n=16,849$) is the 50th class, not shown. Error concentrates on the centre’s immediate neighbours, not distant points.

both colours, so scored per opening: mcts sweeps 20 of 25, splits 5, greedy sweeps *zero*. An exact binomial on sweeps alone (most conservative, discarding the 5 splits) gives $p=4.1 \times 10^{-3}$. The ≈ 400 -Elo gap between search and one-shot argmax on identical weights exceeds every supervised-learning effect above.

A first version without opening randomisation reported mcts 50–0: since both agents are deterministic (argmax; no exploration noise), those 50 games were 2 distinct games replayed 25 times each. Discarded, reported here only with randomised openings — a pitfall worth flagging for any deterministic game-playing evaluation, not specific to this corpus.

F. Duplication ablation

§III-E leaves open whether the CNN’s data-volume gain is genuinely more information or just more exposure to repeated opening rows. Retraining the champion cell with `dedup=unique` (56,798 rows for 3,310,156) at the same 12 epochs costs 0.223 top-1 (0.9040 $\rightarrow$ 0.6812) — large enough that Table III’s Factor B result could be an artefact of opening rows dominating gradient steps. Extending to 100 epochs ($58 \times$ fewer steps/epoch) closes 89% of that gap (residual 0.024) — not catch-up, since its training loss (0.204) is already *lower* than the 12-epoch raw model’s (0.383) while validation top-1 stays below it: overfitting a uniform training distribution evaluated against a frequency-weighted one. We read Factor B as dominated by gradient-step budget, with a smaller residual gap from train/val distribution matching, not duplication itself.

G. Unsupervised representation

At latent 64 the autoencoder reaches near-lossless reconstruction (holdout BCE 1.0×10^{-7} , train/holdout tracking together); at 32 it plateaus five orders of magnitude higher (0.0139) — a real bottleneck. Table V shows the two families diverging sharply on linear probes (control $R^2 \in [-0.02, 0.00]$)

TABLE V
LINEAR-PROBE R^2 RECOVERING BOARD STATISTICS FROM A 64-D LATENT. EVERY CONTROL R^2 (SHUFFLED TARGET, IDENTICAL PROTOCOL) IS IN $[-0.02, 0.00]$; EVERY PROBE BELOW CLEARS IT.

Statistic	AE ($L=64$)	AE ($L=32$)	Champion ($N=4$)
Stone count	0.979	0.943	0.268
Move number	0.825	0.801	0.501
Largest group	0.479	0.480	0.155
Group count	0.161	0.052	0.108
Groups in atari	0.087	0.036	0.062
Edge-vs-centre	0.849	0.814	0.271

throughout, omitted for space; every probe clears it): the autoencoder recovers simple global statistics almost perfectly, the supervised representation far more weakly — reconstruction cannot discard occupancy detail without cost; a value head can discard whatever does not help predict the outcome, and does.

Cluster geometry inverts this: the champion has a clean silhouette optimum at $k=3$ (0.468; one 93.0% cluster plus two small late-game clusters, mean move number ≈ 54 vs. 30.6 overall), while no autoencoder run (2 latents $\times$ 3 seeds) ever finds an interior peak — silhouette still rises at $k=20$, our ceiling, at a best value (0.16) a third of the champion’s. Assignment is nonetheless fairly stable across seeds (adjusted Rand 0.73–0.83); reproducible, just not well-separated. DB-SCAN agrees (16 clusters/38% noise for the champion vs. 101–120 fragments per autoencoder run) on a PCA space needing more components for 90% variance (32–33 of 64 vs. the champion’s 22) — supervision compresses to lower effective dimensionality here.

Copying the latent-64 encoder’s three conv layers into a fresh $N=2$ CNN, trained at the smallest volume (1,000 games, 5 seeds, both arms local to avoid §VI’s GPU/CPU drift) gives a positive but inconclusive edge over random init ($0.8280 \pm 0.0058 \rightarrow 0.8334 \pm 0.0035$ top-1, $t(4)=1.55$, $p=0.20$; $0.7554 \pm 0.0370 \rightarrow 0.8076 \pm 0.0576$ value accuracy, $t(4)=2.03$, $p=0.11$) — one row, not a second headline.

VI. DISCUSSION

The CNN-vs-classical comparison is confounded by architecture, not just capacity. A linear model over flattened pixels cannot represent a translation-equivariant filter regardless of data, so Table III and Fig. 2 should be read as *feature richness vs. architecture-appropriate capacity*, not convolution being unconditionally superior: §V-B shows the same planes are worth a large, real amount to linear models precisely because they lack the inductive bias to derive that structure themselves.

The crossover’s dynamic range is bounded by the corpus. $108 \times$ more games buys only $10.9 \times$ more unique positions (§III-E), so Factor B spans roughly one order of magnitude in the quantity that matters; the CNN was still gaining at the last data point (Table III), so the reported crossover is a lower bound, not an observed plateau.

Reproducibility has two distinct failure modes, not one. The corpus is not bit-reproducible on regeneration (KataGo

seeds per run), so exact replication needs our released HDF5 files, not just our generation config. Separately, GPU training is not bit-reproducible from identical seeds and code: the released checkpoint measures 90.40% top-1 against 90.46% for the sweep cell it was re-trained from (same seed, same hyperparameters). Any number here should be read with a ± 0.06 -point band from this source alone, on top of our confidence intervals’ seed-to-seed spread.

The teacher bounds both what “move prediction” can mean and how noisy our playing-strength instrument is. Every label comes from one KataGo network at `maxVisits=16`; a 90% top-1 model imitates that policy well, not optimal 7×7 play, and a stronger teacher would raise every model’s ceiling here. Separately, §III-D shows 71% of self-play games share one outcome; colour-balanced round-robin (§IV-A) mitigates this, but head-to-head play between two *weak* agents still varies more than KataGo’s own strong self-play — our search agent lost 5 of 50 games to one-shot argmax despite a ≈ 400 -Elo gap, a noisy instrument, not a broken one.

Reconstruction and supervision preserve different information, not a strictly better or worse representation. §V-G’s autoencoder recovers raw board statistics the same-dimensional supervised representation does not, while that representation clusters far more cleanly. With §V-B’s finding that the CNN needs no hand-engineered liberty or history planes, both results describe the same architecture from opposite directions: given a target, it discards detail that target does not need; given none, it keeps everything it can.

Two properties of the corpus are disclosed rather than fixed (§III-E): opening-move determinism (only 17 distinct positions exist by ply 4) and the resulting 90.7% train/val position overlap. Neither is leakage — game-level overlap is exactly zero — but both compress diversity in the low-ply region, which is also where Factor A’s planes carry the least information; some of the small-volume regime’s flatness for classical models may owe to this, not capacity alone.

VII. CONCLUSION

This paper set out to ask whether hand-engineered Go feature planes still earn their keep as labelled data grows, on a controlled, oracle-verified 7×7 corpus. The answer is architecture-dependent rather than uniform: liberty and history planes buy a linear model up to +0.11 top-1 accuracy and buy a three-layer CNN nothing measurable, an interaction confirmed by a two-way ANOVA on encoding $\times$ model ($p < 10^{-60}$ on the interaction term), not just an effect-size argument. The same CNN reaches 90.4% top-1 move prediction unaided on the sealed test split, overtakes every classical baseline once training data exceeds $\sim 13k$ unique positions, and gains substantially more from search over its own frozen weights (≈ 400 Elo) than from any supervised-learning design choice in the study. A compute-cheap unsupervised sub-study asked a parallel question of the same architecture: a reconstruction-trained autoencoder recovers raw board statistics a same-dimensional supervised representation

does not, while that supervised representation organises into far cleaner clusters — reconstruction and supervision preserve different information about the same board, not one subsuming the other.

Three stated limitations bound how far this generalises: every label comes from one fixed KataGo teacher, so the study measures imitation of one policy, not optimal play; opening-move determinism compresses diversity exactly where Factor A’s planes carry least information; and the AE-init transfer result (§V-G) is directionally positive but underpowered at five seeds. Future work, in priority order: a higher-`maxVisits` teacher to raise every model’s ceiling here; randomised opening plies during generation, decoupling genuine diversity from the opening-book artifact; and a larger- n transfer-ablation re-run, deliberately scoped here as one row, not a second headline.

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AI USAGE STATEMENT

Generative AI (Claude, Anthropic) was used for code debugging, concept explanation, and proofreading, per Guidelines v1.1’s permitted uses; it did not write the manuscript prose. All analysis, interpretation, and writing decisions are the authors’ own. A usage log is available on request.

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